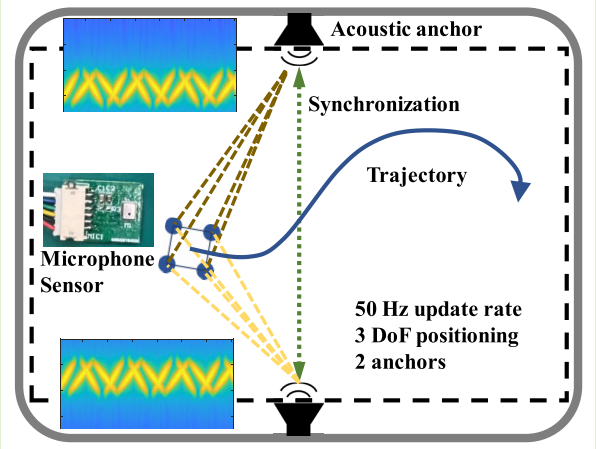
HARP: 3-DoF High-Update-Rate Acoustic Rigid-Body Positioning

Yuzhang Xi, Weimeng Cui, Guangyao Liu, and Zhi Wang, *Member, IEEE*

***Abstract*—Acoustic indoor positioning has been widely studied due to its advantages of low cost, easy deployment, and high accuracy. However, the limitation of insufficient update rates remains a challenge for its application in mobile target localization and navigation. To tackle the obstacles, this article presents high-update-rate acoustic rigid-body positioning (HARP). HARP employs a combined frequency-division multiple access (FDMA) and code-division multiple access (CDMA) signal scheme, achieving stable signal detection at 50 Hz based on an optimal pairing initialization process and a constraint-enhanced cross correlation (CE-CC) detection algorithm. Furthermore, HARP is capable of achieving three-degree-of-freedom (3-DoF) estimation, including planar position and target heading, using only two anchors. HARP updates the initial position (IP) using three-point collinearity rigid-body localization (TPC-RBL) and converts time difference of arrival (TDOA) into time-of-arrival (TOA) observations via continuous distance increment (CDI), enabling localization without synchronization between anchors and sensors. Real-world experiments validate the system's signal detection capability and localization accuracy. HARP achieves a 50-Hz update rate, significantly surpassing existing acoustic systems, while attaining an average localization accuracy of 38.8 mm and an angular accuracy of 6.42° within a room-scale area. This system provides a promising solution for low-cost indoor rigid-body localization (RBL) of mobile targets.**



***Index Terms*— Acoustic indoor positioning, high update rate, rigid-body localization (RBL), signal detection, time difference of arrival (TDOA).**

I. Introduction

INDOOR positioning systems are attracting increasing research and demand with the development of urban construction and intelligent technologies. Compared to global navigation satellite system (GNSS)-based positioning solutions that dominate outdoor environments, indoor positioning offers a variety of solutions such as acoustics [1], ultrawideband (UWB) [2], Bluetooth [3], Wi-Fi [4], 5G [5], and ultrasonic [6]. Each of these approaches has its own advantages and limitations, making them suitable for different scenarios and applications [7].

Among these, acoustic methods offer significant advantages, including low cost, high accuracy, and wide coverage [8]. As a result, acoustic positioning has already been successfully implemented in systems and practical applications for pedestrian and mobile phone localization and navigation among diverse scenarios [9], [10], [11], [12].

However, acoustic localization methods are rarely applied to mobile platforms such as robots, autonomous ground vehicles (AGVs), and autonomous aerial vehicles (AAVs). This limitation primarily arises from two challenges: 1) Doppler shift and 2) low update rates. Autonomous navigation of mobile platforms typically relies on vision [13] or LiDAR [14], [15], which offer update rates of 10–100 Hz and centimeter-level accuracy. However, due to constraints such as weight, power consumption, computational resources, cost, and privacy concerns, these technologies are not suitable for all scenarios. Passive localization is often based on an anchor–sensor framework. UWB technology is immune to Doppler shift and provides update rates exceeding 50 Hz with centimeter-level accuracy [16], making it a valid solution for mobile target localization. Similarly, ultrasonic localization systems can achieve a 100-Hz update rate with centimeter-level positioning accuracy [6]. Acoustic localization can also achieve submeter or centimeter-level accuracy. However, the accuracy

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degradation caused by the Doppler shift and the limited applicability due to low update rates prevent it from being a viable solution for mobile platforms.

The acoustic positioning errors caused by the Doppler shift due to mobile targets cannot be overlooked. Many studies have focused on eliminating the errors and have achieved satisfactory results, including explicit compensation methods based on velocity estimation [17] and those requiring no velocity information approaches implicitly eliminating errors with signal processing [18], [19].

Mobile targets typically require a high update rate to ensure trajectory accuracy and real-time navigation. However, most acoustic localization methods have an update rate of only 1–2 Hz [1]. The low update rate issue of acoustic methods remains an insurmountable obstacle. In terms of improving update rates, the limitation specifically manifests in the following three aspects.

1) *Limited Bandwidth:* The mainstream acoustic positioning is based on a four-anchor model. Unlike UWB or ultrasonic methods, the bandwidth of acoustic positioning is extremely limited (16–24 kHz), making it hard to directly utilize frequency-division multiple access (FDMA) technology to distinguish multiple anchors. Acoustic positioning introduces the anchor guard interval (AGI) and sends positioning signals at different times, sacrificing real-time performance and update rate.
2) *Signal Detection Issues:* In the absence of FDMA, code-division multiple access (CDMA) is a potential solution. For instance, using up-down chirps and introducing functions for differentiation can increase the update rate [10]. However, this improvement is still quite limited. When the update rate increases further, distinguishing different encoded signals will become a major problem.
3) *Anchor Delay and Period Guard Intervals:* The existing anchor layout results in anchor delays during signal propagation, which not only affect positioning accuracy but also lead to the well-known "late emitting, early arrival" problem. Period guard intervals (PGIs) address this issue but prevent an increase in the update rate.

In this article, we propose high-update-rate acoustic rigid-body positioning (HARP). This novel positioning system significantly mitigates the impacts of the aforementioned issues. The key innovations and contributions of this work are summarized as follows.

1) We introduce a unique rigid-body localization (RBL) method based on the detection and compensation of three-point collinearity (TPC), enabling the localization of a two-anchor and a three-degree-of-freedom (3-DoF) under specific conditions. Building upon this, we utilize continuous distance increment (CDI) calculation and initial position (IP) iteration to achieve continuous trajectory localization with no cumulative errors.
2) We design a signal scheme combining FDMA and CDMA, achieving an update rate of 50 Hz, a value far exceeding that of existing systems.
3) To support the signal detection for this complex scheme, we design an optimal pairing-based signal detection initialization process and a constraint-enhanced cross correlation (CE-CC) detection algorithm.
4) We implement the HARP system, and real-world experiments demonstrate that this low-cost system can achieve high-precision 3-DoF RBL, including 2-D planar position and target heading. In a room-scale testing area, the mean positioning error is 0.0388 m and the yaw error is 6.4211°, offering a significant practical value.

The remainder of this article is organized as follows. Section II introduces the related works of this article. Section III describes the overall workflow and deployment plan of the system. Section IV provides a detailed introduction to the system's signal design scheme and detection methods. Section V introduces the system's positioning process and algorithms. Section VI presents the experimental results, and Section VII summarizes the article and outlines future work.

II. Related Works

A. Rigid Body Positioning

RBL can provide rotation and position information and is widely applied in mobile target positioning. In recent years, the RBL field has been deeply studied. According to the basic positioning approaches, it can be divided into three categories: angle of arrival (AOA), time of arrival (TOA), and time difference of arrival (TDOA). Specifically, the AOA approach has high accuracy and requires fewer anchors [20], which is most conducive to improving the update rate. However, its positioning range is quite limited and cannot meet the application scenarios of indoor acoustic positioning. For example, [21] and [22] achieved 2-D and 3-D positioning with only one anchor, respectively. The target was approximately 1 m away from the anchor, and the 90th percentile positioning error is 26.5 and 58.3 cm, respectively. In [23], RBL within a 4 m^2 area was achieved, with the 90th percentile of the position errors being less than 11.44 cm and the yaw error being less than 7.96°. In [24], RBL of moving targets was achieved within a 0.6 m^2 area, with the median positioning error of 2.6 cm and the rotation error being less than 10°.

The TOA and TDOA methods are similar. TDOA does not require synchronization between anchors and target sensors. Although it requires more anchors and sensors to ensure positioning accuracy (more than 4), it is easy to expand and can carry multiple targets, thus having a wide range of applications. The most common solution methods for TOA and TDOA are the maximum likelihood (ML) estimation method [25]. However, the ML function is nonconvex and often converges to a local optimum. Chepuri et al. [26] proposed the constrained least-squares (CLS) and simplified CLS (SCLS) methods, but neither of them can achieve the Cramer–Rao lower bound (CRLB). Jiang et al. [27] utilized semi-definite relaxation (SDR) to enable the TOA-based method to stably converge to the CRLB when the noise is small. Based on this, [28] further used SDR to implement RBL based on TDOA measurement.

To summarize, realizing room-scale acoustic RBL systems that simultaneously maintain centimeter-level positioning accuracy while operating with minimal anchor deployment presents a significant technical challenge. Current methodologies struggle to fulfill these stringent criteria,

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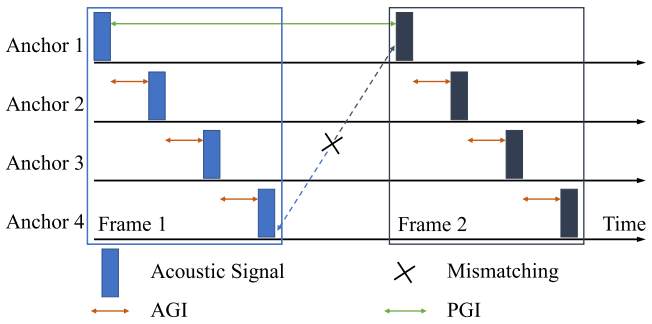


Fig. 1. Signal scheme of traditional acoustic positioning.

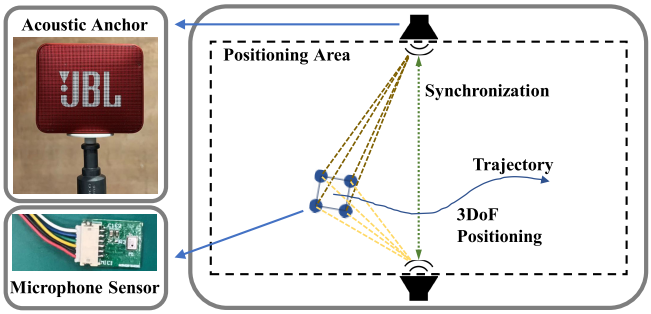


Fig. 2. HARP system layout diagram.

particularly in balancing system complexity with precision requirements.

B. Signal Scheme

The traditional acoustic positioning adopts a positioning scheme with four anchors, usually based on TOA or TDOA positioning, as described in [10], [11], [12], and [29]. Its signal scheme is shown in Fig. 1, with four signals per frame. There is AGI between signals and PGI between frames. AGI is set up to prevent overlapping and interference of signals from different anchors within the same frame due to limited bandwidth; PGI is established to prevent frame-to-frame mismatch caused by signal propagation delay. These two are interrelated, and their existence limits the upper limit of acoustic positioning update rate.

In conclusion, based on the existing anchor layout and signal scheme, the update rate cannot be significantly improved. Therefore, a completely new positioning system needs to be designed.

III. System Overview

The HARP system relies on two position-known and synchronized acoustic anchors to achieve 3-DoF localization of the target microphone array with a 50-Hz update rate. Traditional localization methods place four anchors at the four corners of the localization rectangle. In the HARP system, as shown in Fig. 2, two anchors are placed at the midpoints of the two parallel edges of the localization rectangle. The length of one side of the localization rectangle is determined by the distance between the two anchors, while the length of the other side is mainly determined by the speaker power and sound attenuation, that is, whether the microphone array can still effectively detect the sound signal.

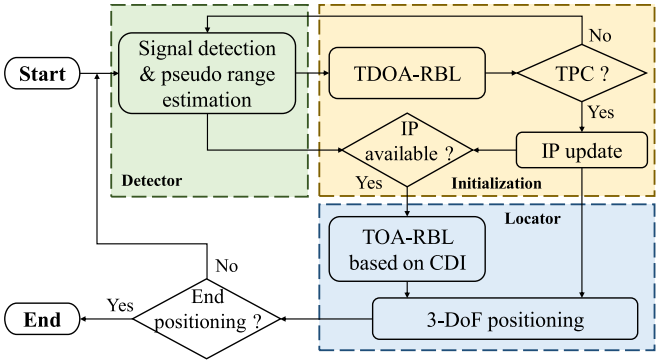


Fig. 3. HARP system flowchart.

The workflow of the HARP system is illustrated in Fig. 3, and it mainly consists of three components. The first is the detector, which performs signal detection and obtains a set of eight pseudorange estimates, as marked by the dashed lines on the right side of Fig. 2. The second is the initialization process. Unlike traditional localization methods, HARP can only start localization when the target triggers the TPC condition and provides an IP. TPC represents the condition where any microphone of the target lies on the line connecting the two anchors. This condition can be triggered unintentionally or deliberately by the moving target crossing the line. During target movement, whenever a TPC event is detected, the target's IP can be updated, reducing the influence of initial localization errors on subsequent positioning. Finally, the locator component takes over. Once an IP is available, HARP uses CDI to convert pseudorange values into real distance measurements, enabling RBL based on TOA. It is noteworthy that this process does not produce cumulative errors; the error in each localization instance is solely related to the current pseudorange estimation error and the initial localization error.

In this article, the microphone array used for target localization consists of four synchronized consumer digital MEMS microphones, each with a size of only 4 × 3 mm. This makes the sensor array inexpensive and easy to deploy on platforms such as mobile robots, carts, helmets, and so on. For distributed wireless sensor networks of indoor positioning systems, time synchronization [30] is one of the key issues. HARP achieves localization independently with only two acoustic anchors, which reduces synchronization overhead. Moreover, unlike traditional methods, when expanding the localization area, the newly added anchors only need to change the transmitted signals and do not need to be synchronized with the old ones.

IV. Signal Scheme and Detection

In this section, we will introduce the signal design scheme and signal detection algorithm of the HARP system, which are essential for achieving the high update rate of 50 Hz.

A. Signal Scheme

To maximize the update rate, the HARP system employs a combination of CDMA and FDMA. FDMA is the simplest and most effective differentiation method. Using bandpass filters, we can focus solely on the received signal in a

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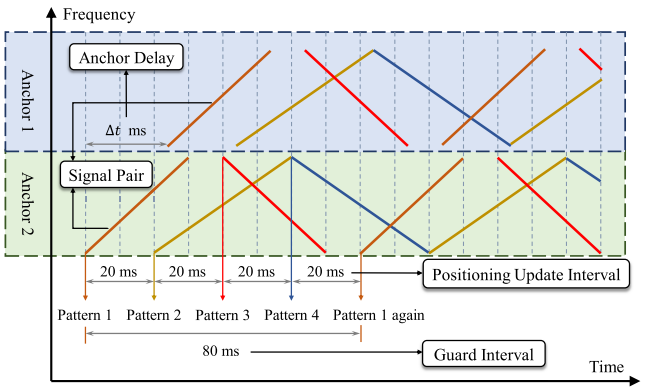


Fig. 4. Received signal diagram of HARP.

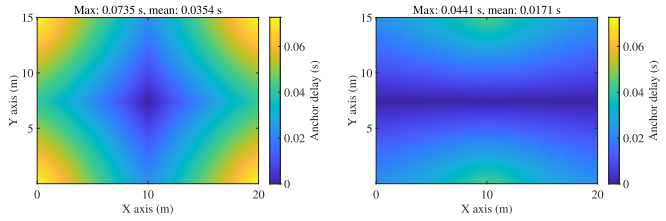


Fig. 5. Example of anchor delay distribution. (a) Layout of four anchors. (b) Layout of HARP.

specific frequency band. Since the target and different anchors could be at varying distances, the received signals from different anchors may have different gains and time delays. To distinguish the signals from the two anchors as much as possible, the HARP system utilizes FDMA at anchors, with the two anchors occupying the 16–19- and 19–22-kHz frequency bands, respectively, as shown in Fig. 4.

A single positioning process requires the detection of a group of signals, with the number of signals in the group equal to the number of anchors. In traditional methods, due to limited available bandwidth, AGI must be maintained, meaning that a group of signals cannot be transmitted simultaneously. In contrast, for the HARP system, a pair of signals can be transmitted at the same time, as shown in Fig. 4. For a signal pair, we use two chirp signals with the same duration and chirp rate, which facilitates pairing during detection. During signal pairing, the principle of nearest pairing is generally adopted. However, there is an anchor delay in the receiving signal, which is denoted as Δt in Fig. 4. When Δt is large, it may lead to an incorrect pairing issue. Therefore, it is necessary to set a PGI, ensuring PGI $>\Delta t$. The HARP system uses an anchor layout different from traditional methods, which reduces the PGI and the average anchor delay. The maximum anchor delay at a specific target location is

$$\Delta t = \left(\max_{m,n} \|\mathbf{a}_n - \mathbf{s}_m\|_2 - \min_{m,n} \|\mathbf{a}_n - \mathbf{s}_m\|_2\right) \Big/ c \quad (1)$$

where c is the sound speed and $\mathbf{a}_n$ and $\mathbf{s}_m$ represent the location of nth anchor and mth sensor, respectively. Taking a 15×20 m localization area as an example, the anchor delays within this area are illustrated in Fig. 5. The maximum anchor delays for the two layouts are 73.5 and 35.4 ms, respectively, while the average anchor delays are about 44.1 and 17.1 ms,

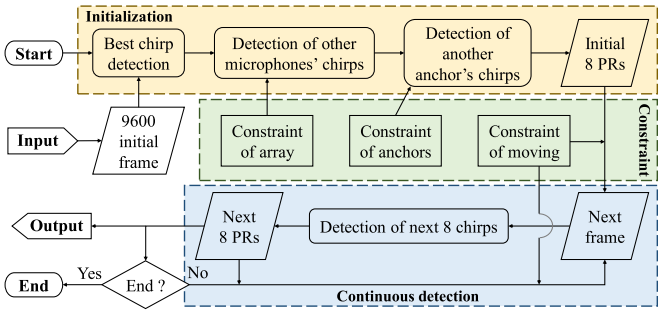


Fig. 6. Signal detection flowchart.

respectively. To achieve a sufficiently large localization range, we set the PGI to 80 ms, corresponding to a maximum anchor distance of 27.2 m.

At this point, the localization update rate reaches 12.5 Hz. To further improve the update rate, we adopt CDMA, transmitting chirp signals with different parameters every 20 ms within a single PGI. Taking anchor 1 as an example, the signal parameters are (19–22 kHz, 30 ms), (19–22 kHz, 40 ms), (22–19 kHz, 30 ms), and (22–19 kHz, 40 ms), respectively. At last, the signal scheme achieves an update rate of 50 Hz.

It can be observed that different chirp signals overlap within a single PGI, which may lead to detection errors. In theory, the cross correlation (CC) algorithm can distinguish between different chirp signals, as infinitely long linear chirp signals form a complete set of orthogonal bases in the fractional Fourier transform domain [31]. However, in practice, the CC results are significantly affected by signal length and gain. In Section IV-B, we will introduce the optimal pairing-based signal detection initialization process of HARP and the proposed CE-CC algorithm.

B. Signal Detection

In this subsection, we will provide a detailed explanation of the signal detection process. The detector in Fig. 3 is specified and its flowchart is shown in Fig. 6. The detector mainly consists of three parts: initialization, constraints, and continuous detection.

In the initialization step, the initial frame length is set to 9600 with a sampling rate of $f_s = 48$ kHz. HARP first performs 32 CCs among the initial frames of the four microphones and eight different reference chirp signals. The chirp corresponding to the maximum peak value of all the CC results is considered the best chirp, and its arrival time is denoted as tbest. In practical processing, all data are in discrete form, with the TOA represented by the sample index ibest, and ibest $=$ tbest $\cdot f_s$. This process can be formulated as

$$\begin{aligned} R_{\max(p,m)} &= \max_i \left\{R_{sr(p,m)}\right\} \\ (\text{pbest}, \text{mbest}) &= \arg\max_{(p,m)} \left\{R_{\max(p,m)}\right\} \\ \text{ibest} &= \arg\max_i \left\{R_{sr(\text{pbest},\text{mbest})}\right\} \end{aligned} \quad (2)$$

where p is the number of different chirp patterns, m is the microphone number, R_{sr} is the CC result, s is the reference signal, r is the received signal, i is the index number in R_{sr},

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$R_{\max}$ is the peak value of each R_{sr}, and pbest and mbest are the corresponding pattern number and microphone number of $R_{\max}$, respectively.

After obtaining the best chirp number and microphone number (pbest, mbest), we use the array constraint to determine the position of the other microphones' corresponding frames. This process can be expressed as

$$\begin{aligned} C_m^{\text{array}} &= \|\mathbf{s}_m - \mathbf{s}_{\text{mbest}}\|_2 \cdot f_s/c \\ \text{st}_m &= \text{ibest} - C_m^{\text{array}} \\ \text{ed}_m &= \text{ibest} + C_m^{\text{array}} + \text{len}\left(s_{\text{pbest}}\right) \end{aligned} \tag{3}$$

where C_m^{array} is the array constraint of microphone m, len$(\cdot)$ represents the signal length, and $[st_m, ed_m]$ is the interval of the determined frame. Then, CE-CC is used to detect the arrival time i_m of the chirp in this frame. The microphone array constraint provides signal arrival window prediction in the initialization step to reduce interference from other signal patterns, noise, and multipath effects.

Similarly, when determining the frame of the signal pair for another anchor, we use the constraint of anchors

$$\begin{aligned} C_m^{\text{anchor}} &= \|\mathbf{a}_1 - \mathbf{a}_2\|_2 \cdot f_s/c \\ st_m^{\text{pair}} &= i_m - C_m^{\text{anchor}} \\ ed_m^{\text{pair}} &= i_m + C_m^{\text{anchor}} + \text{len}\left(s_{\text{pbest}}\right) \end{aligned} \tag{4}$$

where C_m^{anchor} is the anchor constraint of microphone m and $[st_m^{\text{pair}}, ed_m^{\text{pair}}]$ is the corresponding frame. The constraint of anchors arises from the fact that, for any pair of signals simultaneously transmitted from anchors on both sides, the maximum time difference for reception at any microphone must be less than the maximum possible propagation time difference between the two anchors. The anchor constraint is used to provide the predicted arrival window for the corresponding four signals transmitted by another anchor. Its primary purpose is to ensure that the window can fully cover the signals. When the anchor distance is smaller than the maximum distance determined by the PGI (80 ms), this constraint ensures that only one valid target signal appears within the allowed window.

After the initial eight pseudo-range (PR) measurements are completed, HARP enters the continuous detection phase. Unlike traditional methods that use a fixed step sliding window, HARP employs the constraint of moving to predict the next frame, thereby minimizing the interference from other chirp patterns. The process is similar to the one described above

$$\begin{aligned} C^{\text{moving}} &= \lceil T v f_s/c \rceil + T f_s + N_{\text{bias}} \\ st_{\text{next}} &= i_{\text{last}} - C^{\text{moving}} \\ ed_{\text{next}} &= i_{\text{last}} + C^{\text{moving}} + \text{len}\left(s_{\text{next}}\right) \end{aligned} \tag{5}$$

where C^{moving} is the constraint of moving, T is the positioning interval ($T = 20$ ms), v is the top speed of the object, and N_{bias} is the hardware static bias.

By predicting the frame position, we reduce the length of other chirps within the frame. Meanwhile, adjacent chirps, which are emitted from the same anchor, experience minimal displacement of the target in a short period, resulting in similar gains. As a result, the target chirp signals will be more likely to achieve the maximum CC peak.

To further leverage the constraints, we designed the CE-CC algorithm, as shown in Algorithm 1, for robust signal detection. The algorithm is utilized in every detection process of HARP. In Algorithm 1, Hilbert$(\cdot)$ stands for the Hilbert transform operator.

Algorithm 1 CE-CC

Input: reference s, frame r, st, ed
Output: TOA i, $state$

$R_{sr} \leftarrow$ cross correlation s, r ;
$\widehat{R_{sr}} \leftarrow |Hilbert(R_{sr})|$;
$pv_m \leftarrow \max \widehat{R_{sr}}$;
$\overrightarrow{peakValue}, \overrightarrow{peakIndex} \leftarrow$ find peaks from $\widehat{R_{sr}}$;
$temp \leftarrow 0$, $i \leftarrow 0$, state $\leftarrow$ **true** ;
for $j = 1$ **to** $len(\overrightarrow{peakValue})$ **do**
  $pv, pi \leftarrow \overrightarrow{peakValue}[j], \overrightarrow{peakIndex}[j]$;
  if $pv < \lambda \cdot pv_m$ **then**
    $\overrightarrow{peakValue}[j] \leftarrow 0$, **continue**;
  end
  if $pi + len(s) > ed - st + 1$ **or** $pi < 1$ **then**
    $\overrightarrow{peakValue}[j] \leftarrow 0$, **continue**;
  end
  if $pv > temp$ **then**
    $temp \leftarrow pv$, $i \leftarrow pi$;
  end
end
$i \leftarrow i + st$;
if $\sum \overrightarrow{peakValue} = 0$ **then**
  $state \leftarrow$ **false** ;
end

Let s denote the reference signal and r denote one frame of the received signal. The start and end positions of this frame are denoted as st and ed, respectively, which are determined by the output of a constraint module. We perform cross correlation between the reference signal and the received frame, and then take the absolute value of the Hilbert transform of the result to obtain the output $\widehat{R_{sr}}$. Let the maximum value of the output be denoted as pv_m. CE-CC extracts all peaks from $\widehat{R_{sr}}$ along with their indices, denoted as vectors $\overrightarrow{\text{peakValue}}$ and $\overrightarrow{\text{peakIndex}}$, respectively. These peaks and indices are then individually evaluated. In Algorithm 1, pv represents a single candidate peak value and pi denotes its corresponding index. The algorithm first uses a low peak threshold λ (λ is 0.1 in this article) to eliminate severe noise, multipath, and interference from other encoded signals. Then, it leverages a predicted signal arrival window to reduce signal loss caused by target movement, while simultaneously suppressing interference from multipath and other coded signals.

Among the candidates that satisfy both constraints, the one with the highest peak is selected, and its corresponding index is taken as the estimated TOA i. This iterative process is

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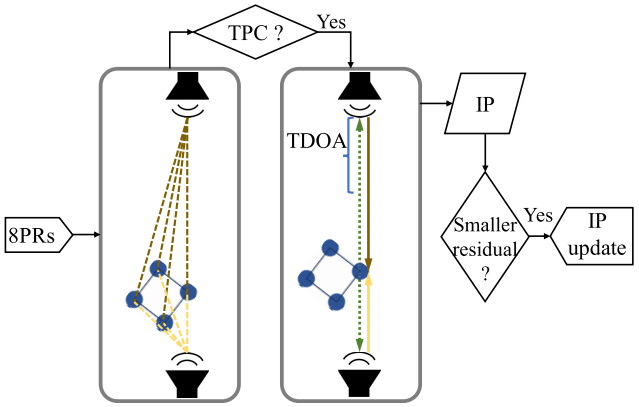


Fig. 7. TPC-RBL illustration and flowchart.

tracked using a temporary variable temp. Finally, if the status flag state is **false**, it means that no valid signal was found under the constraints. It is possibly due to hardware failure, signal blockage, or impulsive noise saturating the microphone. In such a case, the result is discarded as well. And the output time i is then set to st, not as a valid arrival time, but to provide a static constraint for the next estimation.

In summary, CE-CC examines all peaks greater than 0.1 times the maximum value, determines whether the corresponding signals meet the predicted frame constraints, and selects the time of the largest peak that satisfies the constraints as the signal arrival time.

V. Positioning With Two Acoustic Anchors

The premise of the signal scheme is the unique anchor layout of HARP. Therefore, achieving localization with just two anchors is the most critical foundation of the system. In this section, we will demonstrate how to conquer this challenge.

A. TDOA-RBL With TPC Compensation

The detailed process of TPC-RBL is shown in Fig. 7. This step marks the beginning of the localization process. As shown in the system flowchart Fig. 3, if the TPC condition is not satisfied, the system will proceed to detect the next set of signals and once again determine whether the TPC condition is met. This means that localization can only begin after the microphone passes through the line connecting the two anchors. Once the TPC condition is satisfied, the system can continue localization without requiring the microphone to achieve TPC again. Furthermore, if the microphone satisfies the TPC condition again and successfully updates the IP, it will help to reduce the localization error. The triggering of TPC occurs when the moving target passes through the connection line between two anchors, either unintentionally or through deliberate control. Therefore, when deploying the anchors, it is important to ensure that the target has an opportunity to pass through this connection line. Section V-A describes the detailed implementation of this process.

First, we obtain a naive RBL with eight PR measurements using the ML estimation (MLE) method based on the TDOA observations. Assume that the mth sensor position in the local reference frame is $\mathbf{c}_m$. The global coordinates of the mth sensor are denoted as $\mathbf{s}_m = \mathbf{R}\mathbf{c}_m + \mathbf{t}$. Here, when all height information is known, $\mathbf{R}$ is the rotation matrix which belongs to the special orthogonal group $\mathrm{SO}(2) = \{\mathbf{R} \in \mathbb{R}^{2\times 2}, \mathbf{R}\mathbf{R}^T = \mathbf{I}, \det(\mathbf{R}) = 1\}$ and $\mathbf{t}$ is the translation vector. The TDOA value can be denoted as

$$\hat{r}_{m,n} = \left\| \mathbf{a}_n - \left(\hat{\mathbf{R}}\mathbf{c}_m + \hat{\mathbf{t}}\right) \right\|_2 - \left\| \mathbf{a}_n - \left(\hat{\mathbf{R}}\mathbf{c}_1 + \hat{\mathbf{t}}\right) \right\|_2. \tag{6}$$

Then, the MLE can be formulated as

$$\mathbf{P}\left(\Delta\mathrm{PR}_{m,n}|\mathbf{s}\right) = \prod_{m,n}^{M,N} \frac{1}{\sqrt{2\pi}\sigma} \exp - \frac{\left(\hat{r}_{m,n} - \Delta\mathrm{PR}_{m,n}\right)^2}{2\sigma^2}. \tag{7}$$

And the naive estimate of RBL is

$$\begin{aligned} \hat{\mathbf{R}}, \hat{\mathbf{t}} &= \arg\max_{\mathbf{R},\mathbf{t}} \ln \mathbf{P}\left(\Delta\mathrm{PR}_{m,n}|\mathbf{s}\right) \\ &= \arg\min_{\mathbf{R},\mathbf{t}} \sum_{m,n}^{M,N} \left(\hat{r}_{m,n} - \Delta\mathrm{PR}_{m,n}\right)^2 \\ \hat{\mathbf{s}}^- &= \hat{\mathbf{R}}\mathbf{c} + \hat{\mathbf{t}}. \end{aligned} \tag{8}$$

It is worth noting that in (6), the two terms in $\hat{r}_{m,n}$ share the same anchor index n. This indicates that among the seven TDOA measurements derivable from the observed eight PRs, we use only six for the MLE, compared to traditional methods. Moreover, all values represent time differences relative to the same anchor, and the TDOAs between different anchors are not utilized. The reason for this design is that, due to the limited number of anchors and their irregular deployment, the TDOA caused by different distances to different anchors is much larger than the TDOA caused by small-array rotations. As a result, traditional algorithms that rely on all TDOAs tend to achieve higher accuracy only along the Y-axis (assuming the anchor-connection line is aligned with the Y-axis), while the accuracy along the X-axis and in heading estimation remains poor. Furthermore, when there are certain errors in the observed TDOAs between anchors, traditional algorithms often severely compromise the accuracy in other degrees of freedom to correct these errors, resulting in significant localization errors or even failure to converge. Therefore, our naive estimate uses only six TDOA constraints. The last constraint between anchors is applied separately during the compensation step, after achieving relatively accurate estimates of the X-axis and yaw. Finally, to directly derive the Y-axis value using anchor constraints, we further leverage the special condition of TPC.

Next, HARP performs TPC detection on the naive estimation, and when the conditions are met, it uses the TDOA values from the collinear microphone to provide a corrected estimation. The condition can be formulated as

$$|\hat{x}^-_{m_0}| \le \epsilon, \quad m_0 = \arg\min_m |\hat{x}^-_m| \tag{9}$$

where $\hat{x}^-_m$ is the x-coordinate of $\hat{\mathbf{s}}^-_m$ and ϵ is a manually set constant. In this case, the m_0 array element is considered collinear with the two anchors, and its coordinates can be calculated as

$$\begin{cases} \hat{x}_{m_0} = 0 \\ \hat{y}_{m_0} = \left(\|\mathbf{a}_2 - \mathbf{a}_1\|_2 - \left(\mathrm{PR}_{m_0,2} - \mathrm{PR}_{m_0,1}\right)\right)/2 + \mathbf{a}_1\,(2)\,. \end{cases} \tag{10}$$

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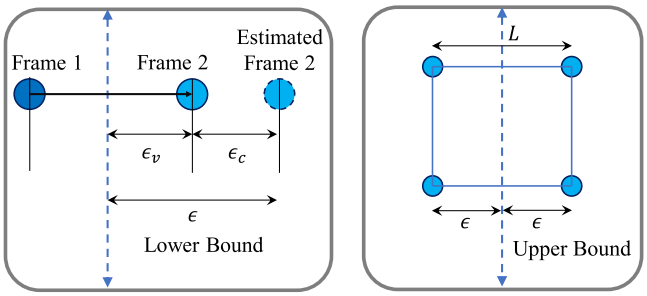


Fig. 8. Value range of the TPC trigger threshold ϵ. (a) Lower bound. (b) Upper bound.

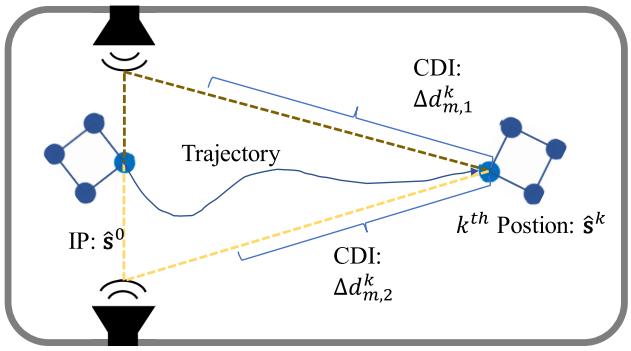


Fig. 9. TOA-RBL based on CDI.

Then, the estimate of IP is

$$\hat{\mathbf{s}} = \hat{\mathbf{R}}\mathbf{c} + \hat{\mathbf{t}} + \hat{\mathbf{s}}_{m_0} - \hat{\mathbf{s}}^-_{m_0}. \tag{11}$$

At this point, the complete IP estimation is achieved. However, the basis of this step is the assumption that when all estimated x-coordinates are less than a threshold ϵ, the microphone is considered to be located on the connection line. As shown in Fig. 8, the threshold ϵ should ensure that the microphone triggers the TPC when passing through the connection line, while also avoiding false detections. Therefore, the following conditions must be met. First, $\epsilon > \epsilon_v + \epsilon_c$, where ϵ_v is the maximum offset caused by target motion and ϵ_c is the maximum reasonable offset caused by CC-based distance estimation. Specifically, $\epsilon_v = v \cdot T/2$ and ϵ_c is determined by the noise conditions. Second, $e < L/2$, where L is the spacing between microphone elements.

Not all estimates satisfying the TPC condition are immediately accepted as the IP value. As a result, we design a residual based on the difference between the observed distance corrected by TPC and the estimated distance. This residual is used to control the IP iteration, ensuring that the estimation error decreases progressively during the iterative process.

Thus, the estimate of distance between microphones and anchors can be formulated as $\hat{d}_{m,n} = \|\mathbf{a}_n - \hat{\mathbf{s}}_m\|_2$. Based on the mth array element, the observed PR measurements can be converted into distance information, which can be calculated by

$$\tilde{d}_{m,n} = \mathrm{PR}_{m,n} - \left(\mathrm{PR}_{m_0,n} - \hat{d}_{m_0,n}\right). \tag{12}$$

We define the residual between distance estimation and observation as

$$\delta = \frac{1}{MN} \sum_{m,n}^{M,N} |\hat{d}_{m,n} - \tilde{d}_{m,n}|. \tag{13}$$

As illustrated in Fig. 7, the IP is updated only when the current residual becomes smaller than the previous estimation. This guarantees monotonic error reduction of IP during TPC-RBL iterations, ensuring progressive enhancement in subsequent positioning accuracy.

B. TOA-RBL Based on CDI

As illustrated in Fig. 3, the HARP locator utilizes TOA-RBL based on CDI, which is shown in Fig. 9. Let the current IP be denoted as $\hat{\mathbf{s}}^0$ and let the estimated position corresponding to the kth captured signal pair be denoted as $\hat{\mathbf{s}}^k$. Then, the CDI between the microphone m and the anchor n can be expressed as

$$\Delta d^k_{m,n} = \left(t^k_{m,n} - t^0_{m,n} - kT\right) \times c \tag{14}$$

where $t^k_{m,n}$ is the arrival time of the kth signal, T is the positioning update interval 20 ms, and c is the sound speed. And the corresponding range measurement is

$$\tilde{d}^k_{m,n} = \hat{d}^0_{m,n} + \Delta d^k_{m,n}. \tag{15}$$

Similar to Section V-A, we continue to employ MLE for RBL. The distinction lies in the adoption of a TOA-based estimation scheme, where the TOA value is modeled as

$$\hat{d}^k_{m,n} = \left\| \mathbf{a}_n - \left(\hat{\mathbf{R}}\mathbf{c}_m + \hat{\mathbf{t}}\right) \right\|_2. \tag{16}$$

The MLE is formulated as

$$\mathbf{P}\left(\tilde{d}^k_{m,n} | \mathbf{s}^k\right) = \prod_{m,n}^{M,N} \frac{1}{\sqrt{2\pi}\sigma} \exp - \frac{\left(\hat{d}^k_{m,n} - \tilde{d}^k_{m,n}\right)^2}{2\sigma^2}. \tag{17}$$

And the estimate of RBL is

$$\begin{aligned} \hat{\mathbf{R}}, \hat{\mathbf{t}} &= \arg\max_{\mathbf{R},\mathbf{t}} \ln \mathbf{P}\left(\tilde{d}^k_{m,n} | \mathbf{s}^k\right) \\ &= \arg\min_{\mathbf{R},\mathbf{t}} \sum_{m,n}^{M,N} \left(\hat{d}^k_{m,n} - \tilde{d}^k_{m,n}\right)^2 \\ \hat{\mathbf{s}}^k &= \hat{\mathbf{R}}\mathbf{c} + \hat{\mathbf{t}}. \end{aligned} \tag{18}$$

VI. Experiment

We have built a prototype of the HARP system for performance evaluation. We installed a microphone array consisting of four microelectromechanical system (MEMS) silicon microphones on a mobile platform, with dimensions of 39.1 × 39.1 cm; meanwhile, we used two synchronized JBL GO2 Bluetooth speakers with a rated power of only 3.1 W as anchors. To analyze the performance of the rigid body positioning system with a 50-Hz high update rate, we use the NOKOV motion capture system to provide the ground truth.

In the experiments, the maximum speed achievable by the robot platform is $v = 2.4$ m/s, and the localization interval is $T = 0.02$ s, resulting in $\epsilon_v = 0.024$ m. Generally, when the signal-to-noise ratio (SNR) is greater than -10 dB, for

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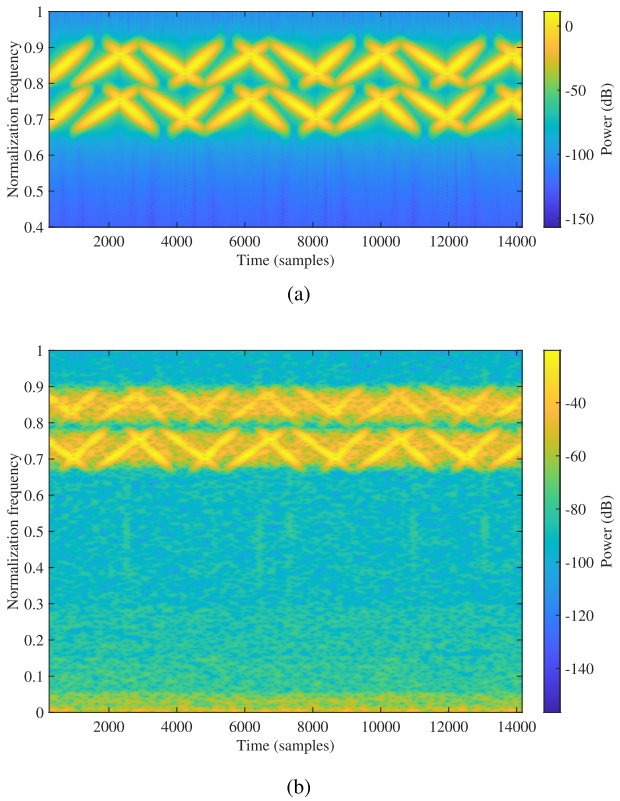


Fig. 10. Spectrogram. (a) Transmitted signal. (b) Received signal.

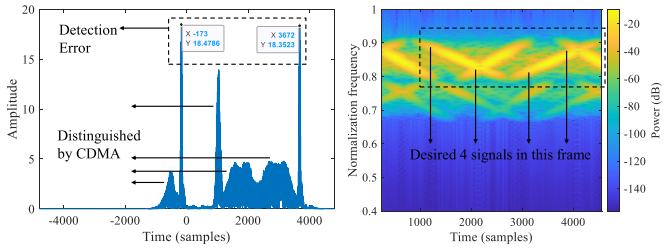


Fig. 11. CC result and a 100-ms frame at 20 s.

chirps with a 3-kHz bandwidth, the accuracy of CC remains within 0.02 m [32], so $\epsilon_c \approx 0.02$ m is used. The microphone element spacing is $L = 0.391$ m. Therefore, the value of ϵ should lie approximately within the range (0.044, 0.196) m. In all experiments, ϵ is set to 0.05 m.

A. Signal Detection

Effective signal detection and TOA/TDOA estimation form the foundation of localization. Severe errors in the estimation can invalidate any localization algorithm. The increased update rate makes robust signal detection extremely challenging. As shown in Fig. 10, the transmitted and received signals with a sampling rate of 48 kHz undergo noise interference and target motion during propagation. The combined effects of CDMA implementation and reduced GI render traditional sliding window detection ineffective.

Fig. 11 displays a frame obtained using a 100-ms window length (4800 samples) with 20-ms overlap (960 samples), captured at the 20-s mark of the received signal. The detection target is a 22–19-kHz chirp signal with 30-ms duration. The CC results on the left demonstrate that CDMA successfully

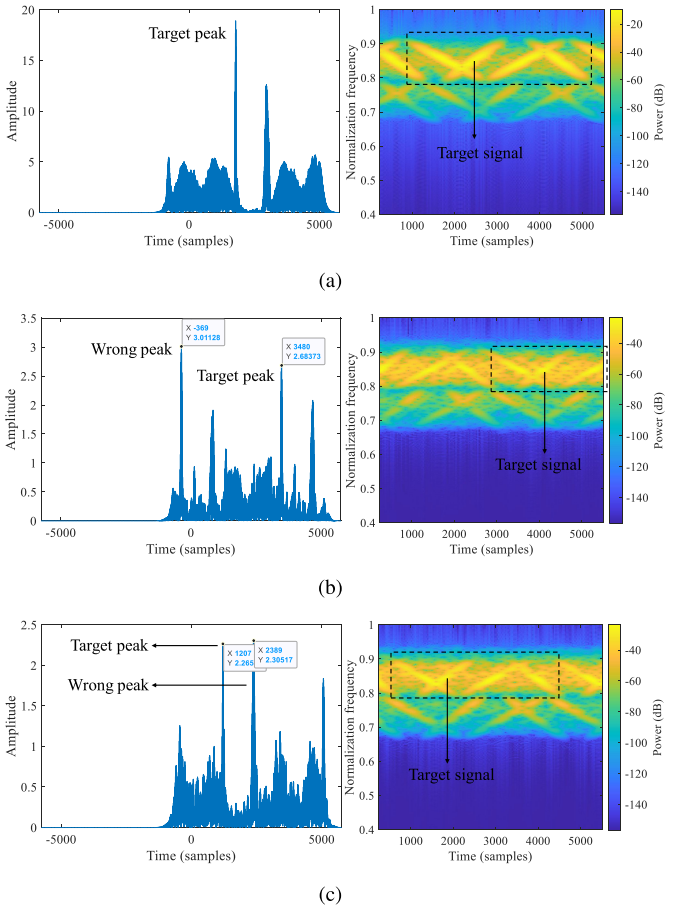


Fig. 12. CC results and 120-ms frames. (a) 20 s. (b) 40 s. (c) 60 s.

distinguishes different encoded signals, yet detection errors persist. A key contributing factor is that rigid body movement, combined with short frame length, causes target signal groups to shift beyond the window boundaries. Thus, previous signals get included in the window while current signals become truncated.

The occurrence of the above problem indicates that the window length is insufficient. Fig. 12 shows the frames obtained using a 120-ms (5760 sampling points) window length and a 40-ms (1920 sampling points) overlap. This window length is equivalent to 1.5 signal repetition periods and six positioning periods. Taking the detection of a 19–22 kHz, 30-ms chirp as an example. As shown in Fig. 12(a), this window successfully covers the target signal set. However, as the rigid body moves, at 40 s, the target signal set again shifts out of the sampling frame. The error in the CC result on the left side of Fig. 12(b) is caused by the previous identical signal. At 60 s, another problem occurs, as shown on the left side of Fig. 12(c). The window completely contains four target signals, but fails to distinguish between 30- and 40-ms signals. This error is difficult to avoid for similar sliding window methods.

Overall, a short window length leads to incomplete signals; a long window length introduces signals from other frames, confusing. Meanwhile, even if the window is long enough, the movement of the target still results in incomplete signal acquisition. Additionally, signal encoding cannot guarantee that the traditional CC algorithm can achieve robust distinction.

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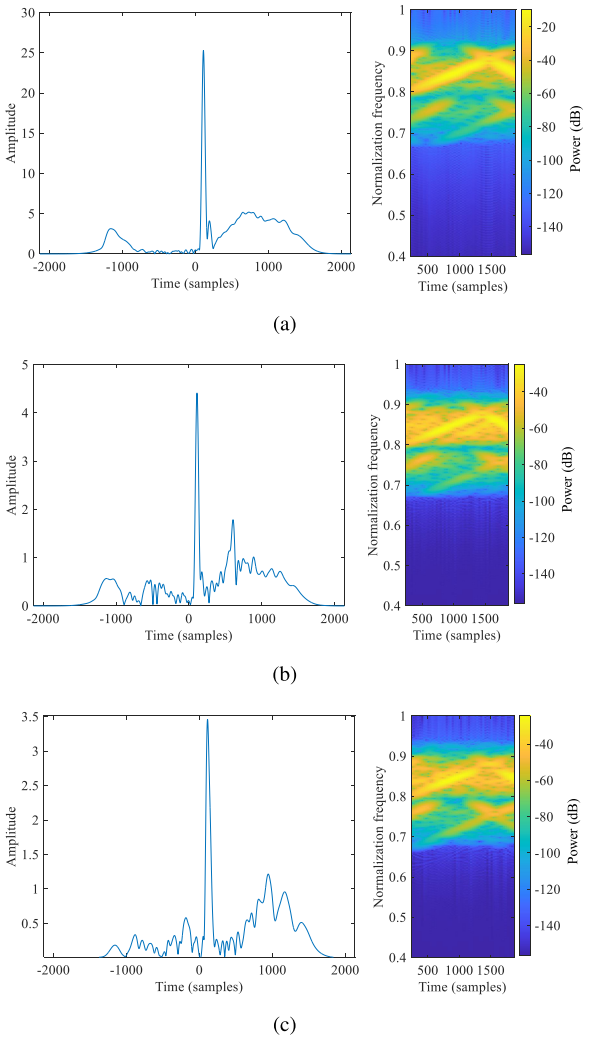


Fig. 13. CC results and frames of the HARP system. (a) 20 s. (b) 40 s. (c) 60 s.

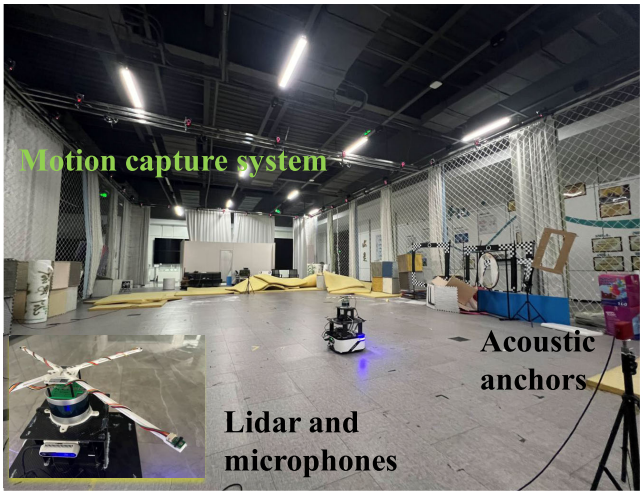


Fig. 14. Experimental setup.

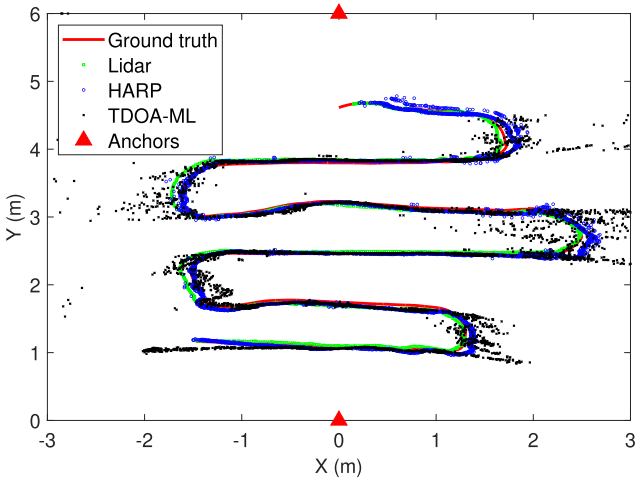


Fig. 15. Trajectory results.

Fig. 13 shows the frames obtained by the HARP system and the CC results at 20, 40, and 60 s, respectively. From the spectrograms on the right side of the three images, it can be seen that the frame acquisition scheme based on motion constraints can always completely obtain the target signal. From the CC results on the left side of the images, it can be seen that the CE-CC method avoids confusion with other codes or multipaths. Even when the cross correlation peak power drops from 25 to 3.5, the detection result remains unique.

B. Analysis of Positioning Accuracy

First, we compare the localization accuracy under identical conditions to demonstrate the superiority of the HARP system's localization scheme. This experiment was conducted on the second floor of the Huzhou Institute of Zhejiang University. The positioning area covered 36 m^2, approximately the size of a normal room, with a distance of 6 m between the two anchors, as shown in Fig. 14. The height of all microphones is 0.652 m, and the height of the two speakers is 0.724 m. As mentioned earlier, a few algorithms can achieve RBL under conditions where the sensors and anchors are not synchronized. In this experiment, we used the proposed algorithm, the traditional TDOA-ML algorithm [26], and the SDR-based algorithm [28]. However, as only two anchors are adopted, the SDR-based algorithm failed to obtain valid positioning results. Therefore, we only show the valid trajectories, as shown in Fig. 15. In this experiment, the TPC parameter of HARP is set as $\epsilon = 0.05$ m. Fig. 16 shows the results of the traditional 4-anchor-based positioning methods. The target velocities under different anchor layouts are shown in Fig. 17, and the cumulative distribution function (cdf) results of the absolute errors are presented accordingly in Fig. 18 and 19.

The localization process of the HARP system initiates only when TPC conditions are satisfied, as evidenced by the trajectory patterns in Fig. 15. The trajectory results reveal a pronounced lateral deviation along the X-axis for the conventional TDOA-ML algorithm. Specifically, Fig. 18 presents the cdf of absolute errors, while Table I provides detailed statistical metrics including mean, median, and 90th percentile errors. As shown in the table, HARP achieves a mean error of merely 0.0169 m, significantly outperforming the TDOA-ML algorithm's 0.5594 m mean error. This performance disparity stems from their distinct measurement frameworks: TDOA-ML employs seven TDOA measurements, whereas HARP integrates six TDOA measurements with eight TOA measurements. The conventional TDOA-ML algorithm exhibits insufficient geometric constraints along the X-axis due to the limited deployment of only two anchors along the

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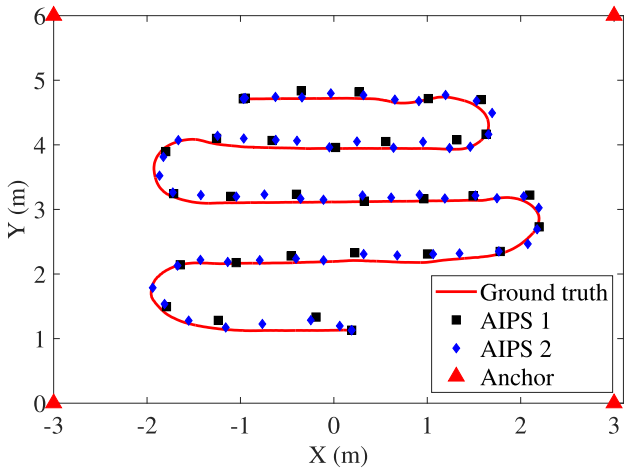
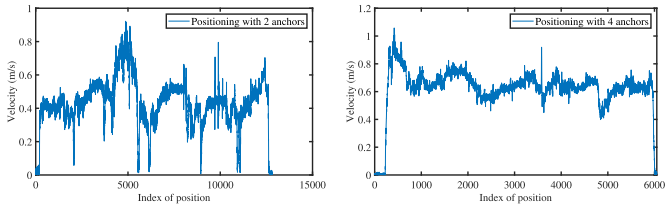
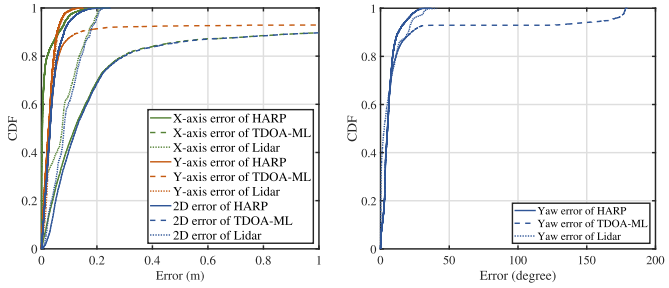


Fig. 16. Trajectory results based on four anchors.



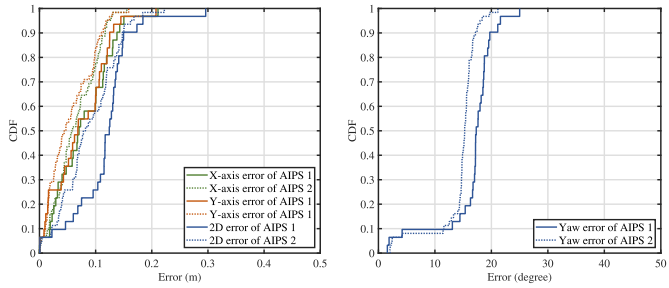
(a) (b)

Fig. 17. Target speeds under different anchor layouts. (a) Positioning with 2 anchors. (b) Positioning with 4 anchors.



(a) (b)

Fig. 18. CDF results of positioning methods (experiment under two anchors). (a) Distance errors. (b) Yaw errors.



(a) (b)

Fig. 19. CDF results of positioning methods (experiment under four anchors). (a) Distance errors. (b) Yaw errors.

TABLE I
STATISTICAL RESULTS OF RBL ERRORS

Item	Method	Mean	Median 50%	90%
X-axis (m)	HARP	0.0169	0.0034	0.0626
	TDOA-ML	0.5594	0.1206	1.0674
	A-LOAM	0.0765	0.0747	0.1704
	AIPS 1	0.0777	0.0704	0.1308
	AIPS 2	0.0641	0.0580	0.1114
Y-axis (m)	HARP	0.0290	0.0259	0.0563
	TDOA-ML	0.1387	0.0310	0.1393
	A-LOAM	0.0374	0.0310	0.0608
	AIPS 1	0.0727	0.0684	0.1248
	AIPS 2	0.0548	0.0455	0.1068
2D (m)	HARP	0.0388	0.0314	0.0819
	TDOA-ML	0.5816	0.1260	1.0712
	A-LOAM	0.0913	0.0813	0.1767
	AIPS 1	0.1156	0.1243	0.1481
	AIPS 2	0.0879	0.0801	0.1475
Yaw (degree)	HARP	6.4211	5.1110	12.9383
	TDOA-ML	17.6082	4.9542	22.3648
	A-LOAM	6.2268	2.6379	20.2624
	AIPS 1	16.4984	17.3191	19.5224
	AIPS 2	14.4614	15.2684	16.8463

Y-axis, resulting in degraded X-axis performance. In Y-axis positioning, both methods demonstrate relatively low median errors (0.0259 m for HARP and 0.0310 m for TDOA-ML). However, HARP maintains superior robustness with a mean error of 0.0290 m and 90th percentile error of 0.0563 m, substantially lower than TDOA-ML's corresponding values of 0.1387 and 0.1393 m. For planar positioning accuracy, the error characteristics align with those observed along the X-axis. Regarding Yaw angle estimation, HARP achieves a mean error of 6.4211°, contrasting sharply with TDOA-ML's 17.6082° mean error. This discrepancy, evident from both trajectory visualization and cdf analysis, arises from occasional catastrophic failures in TDOA-ML outputs, which is a phenomenon absent in HARP's consistently reliable results. The comparative analysis conclusively demonstrates HARP's enhanced robustness and measurement stability across all evaluation metrics.

We also analyzed the accuracy achieved by LiDAR. Specifically, we employed the well-established A-LOAM algorithm [14] along with a VLP-16 LiDAR sensor. The results indicate that the average accuracy of LiDAR is also at the centimeter level. The angular accuracy of LiDAR exhibits a lower mean and median, measuring 6.2268° and 2.6379°, respectively. In this experiment, HARP demonstrates a comparable level of accuracy to LiDAR, while the VLP-16 is priced at around U.S. $300, whereas the HARP system costs less than U.S. $70. For localizing other rigid bodies, additional microphone arrays can simply be incorporated.

In addition, to highlight the differences between other acoustic indoor positioning systems (AIPSs) and HARP, we conducted another trajectory experiment using four acoustic anchors in the environment shown in Fig. 14. The resulting trajectory is illustrated in Fig. 16. The speeds of the target in both experiments were similar, and the corresponding velocity profiles recorded by the motion capture system are shown in Fig. 17. Fig. 19 presents the cdfs of localization errors for the two methods, with detailed error data listed in Table I. It should be noted that we used only the signal schemes and raw localization results of the two methods, without applying

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TABLE II
UPDATE RATE OF EACH METHOD

Method	HARP	TDOA-ML	A-LOAM	AIPS 1	AIPS 2
Update Rate	50 Hz	50 Hz	10 Hz	1 Hz	2 Hz

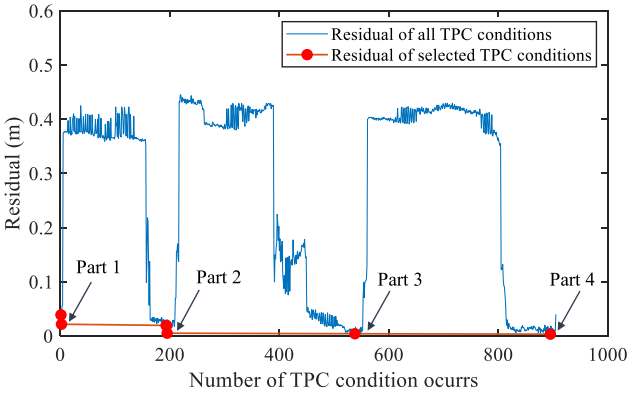


Fig. 20. Residual results of TPC conditions.

the Kalman postprocessing described in the literature. The main difference between the AIPS 1 [12] and AIPS 2 [10] methods is that AIPS 1 includes Doppler compensation but operates at an update rate of 1 Hz, while AIPS 2 does not apply velocity compensation but achieves a 2-Hz update rate. The update rates for all compared methods are summarized in Table II. Both AIPS 1 and AIPS 2 employ four anchors for localization, unlike HARP, and thus exhibit relatively balanced errors along the X- and Y-axes. Taking AIPS 1 as an example, the average errors in the X- and Y-directions are 0.0777 and 0.0727 m, respectively, yielding a ratio of 0.94. In contrast, HARP achieves 0.0169 m in the X-axis and 0.0290 m in the Y-axis, with a ratio of 1.72. Although AIPS 1 and AIPS 2 also attain centimeter-level localization accuracy, their performance is slightly inferior to that of HARP. This discrepancy is primarily attributed to the limited update rate. As discussed previously, the low update rate results from the presence of AGI. Consequently, the target moves a small distance within the AGI, introducing irreducible distance estimation errors across different anchors. This also adversely affects heading estimation, with average heading errors of 16.4984° and 14.4614° for AIPS 1 and AIPS 2, respectively. Moreover, the limited update rate leads to another intuitive distinction between traditional AIPS methods and HARP. As shown in Fig. 16, the trajectory captured by the AIPS methods appears sparse and fails to represent the full path of the target's movement.

C. Analysis of IP Updating

The HARP system derives eight sets of TOA measurements based on CDI. Although free from cumulative errors, (15) demonstrates that each TOA observation remains influenced by current IP estimation results. In this subsection, we investigate whether IP updates progressively reduce estimation errors.

Fig. 20 presents residual errors from 904 TDOA-RBL outcomes under all TPC-satisfied conditions, alongside residuals specifically selected for IP updates. The plot reveals a monotonic reduction in IP residual errors from 0.0388 m during the initial update to 0.0033 m in the final iteration. To further analyze the implications of this residual reduction while excluding near-identical IP updates, we categorize the localization results into four distinct parts. The corresponding outcomes are systematically presented in Fig. 21 and Table III.

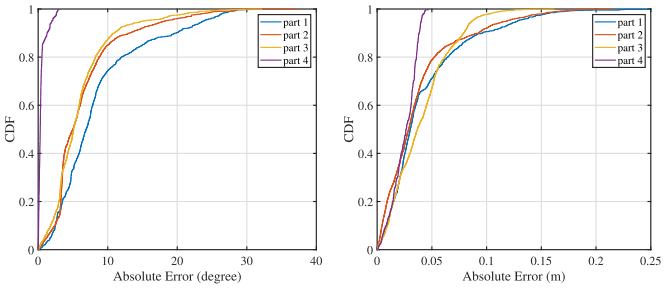


Fig. 21. CDF results of HARP among different parts. (a) Yaw errors. (b) Distance errors.

TABLE III
STATISTICAL RESULTS OF RBL ERRORS AMONG DIFFERENT PARTS

Item	Part	Mean	Median 50%	90%
2D (m)	Part 1	0.0428	0.0301	0.0956
	Part 2	0.0374	0.0287	0.0926
	Part 3	0.0402	0.0368	0.0790
	Part 4	**0.0250**	**0.0264**	**0.0382**
Yaw (degree)	Part 1	8.5751	6.8907	19.6482
	Part 2	6.5080	4.9765	13.0320
	Part 3	6.0277	5.1046	10.8379
	Part 4	**0.4738**	**0.2962**	**1.2754**

In Table III, the smallest values across all metrics are highlighted in bold and exclusively belong to Part 4. Visual inspection of Fig. 21 reveals that post the final IP update, Part 4 demonstrates systematically lower positioning errors compared to the preceding three parts. Statistical analysis in Table III indicates that Part 1 exhibits the largest mean error among the initial three parts, while Parts 2 and 3 show comparable mean and median errors. Notably, a progressive reduction in 90th percentile errors is observed for both 2-D positioning and Yaw angle estimation. This monotonic error attenuation pattern across parts demonstrates that iterative IP updates effectively reduce positioning uncertainties while enhancing system stability.

D. Performance Under Challenging Acoustic Scenarios

1) Noise: The HARP system operates in the near-ultrasound frequency band, and thus its localization performance is affected by ambient noise. To evaluate the system's noise tolerance, we added Gaussian white noise to the signals from the above experiment and performed re-localization. The results under different SNRs are shown in Fig. 22. The original minimum SNR of the received signals was 22 dB. After reducing the SNR, the mean absolute errors (MAEs) are summarized in Table IV. When the SNR drops to 5 dB, the error increases slightly, with a 2-D localization error of 0.0621 m and a heading estimation error of 6.4373°. However, at 0 dB, as illustrated in Fig. 22(d), the system

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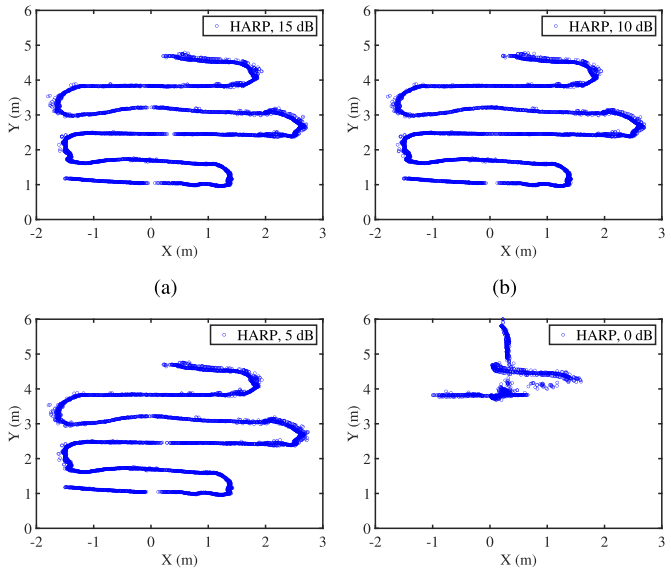


Fig. 22. Trajectories of different SNRs. (a) SNR: 15 dB. (b) SNR: 10 dB.(c) SNR: 5 dB. (d) SNR: 0 dB.

TABLE IV
ERROR DATA UNDER DIFFERENT SNRS

MAE \ SNR	15 dB	10 dB	5 dB	0 dB
2D position (m)	0.0461	0.0496	0.0621	3.1366
Yaw angle (degree)	6.4251	6.4287	6.4373	7.5546

fails completely, indicating that the cross correlation-based signal detection algorithm can no longer provide accurate TOA estimates. This result suggests that the system's operational SNR limit lies between 0 and 5 dB. Therefore, in scenarios with strong background noise in the 16–24-kHz band or significant signal attenuation, increasing the emission power of the anchors can help improve localization accuracy.

2) Multipath: Multipath effects present a major challenge for acoustic localization in indoor environments. We conducted experiments in a room with reflective walls and obstacles such as tables and chairs, as shown in Fig. 23(a). The trajectory in Fig. 23(b) follows a predefined path composed of straight-line segments. Since motion capture systems were unavailable and LiDAR offers comparable accuracy to HARP (thus unsuitable as a reference), we used this predefined path as a substitute for ground truth in performance evaluation.

Under such conditions, although precise error quantification is difficult due to the lack of timestamps, several significant localization errors can be visually identified in Fig. 18(b), particularly near reflective surfaces and obstacles. The most severe error occurs in the upper-right corner, where the target moves into a corner behind tables and chairs. In this case, the signal is affected by both obstruction and multipath reflections from two walls, leading to substantial TOA estimation errors. To provide a coarse error analysis, we calculated the deviation between the estimated positions and the predefined trajectory. As shown in Table V, 2720 points (72.40%) have errors below 5 cm, and 91.06% fall within 10 cm. Notably, 98.64% of the

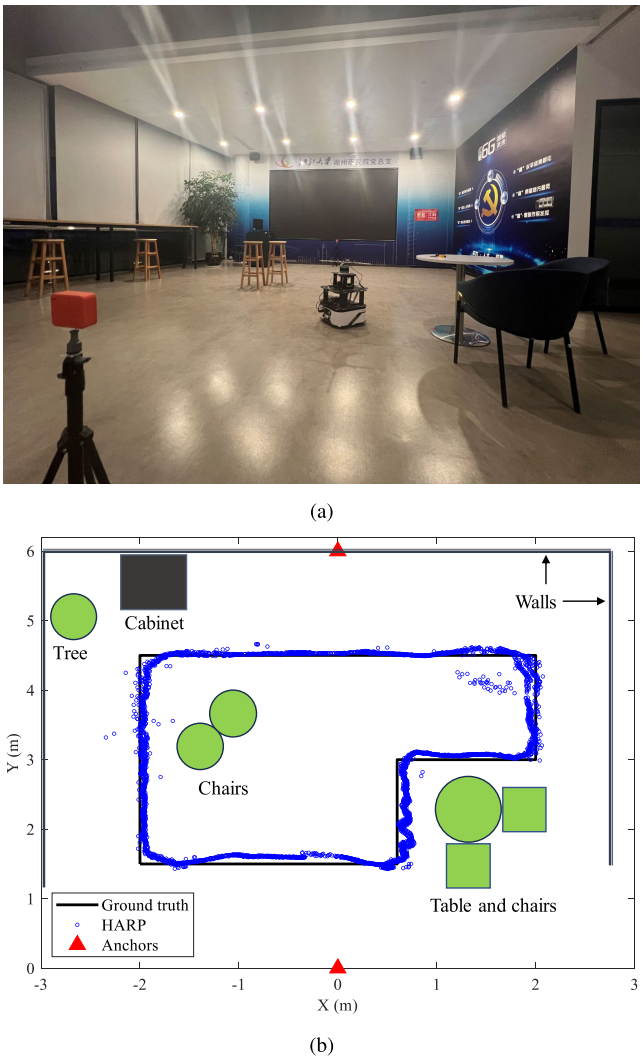


Fig. 23. Experiment in multipath scenario. (a) Setup and scenario. (b) Trajectory.

TABLE V
ERROR DISTRIBUTION IN MULTIPATH ENVIRONMENT

Error	$\leq$ 5 cm	$\leq$ 10 cm	$\leq$ 20 cm
Number	2720	3421	3706
Percentage	72.40%	91.06%	98.64%

errors are below 20 cm. Although slightly degraded compared to open-space performance, this result remains comparable to that of LiDAR systems.

The degradation of localization performance in multipath environments indicates that the signal detection scheme used in HARP still has room for improvement. The CE-CC algorithm in HARP employs constraints based on amplitude and frame length, but it is not specifically designed to address multipath effects. Previous studies [33], [34] have proposed TOA detection algorithms tailored for multipath scenarios. Integrating these algorithms into HARP in future work might enhance the robustness.

3) Doppler Effect: Doppler shifts caused by target motion can introduce errors in CC results. To evaluate the system's tolerance to target speed, we conducted experiments under a

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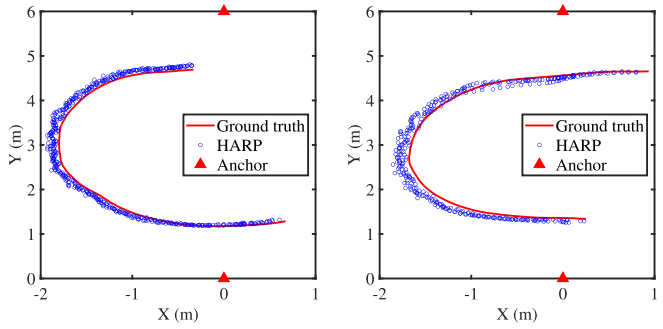


Fig. 24. Two trajectories at different mean velocities. (a) Mean velocity: 1.24 m/s. (b) Mean velocity: 1.97 m/s.

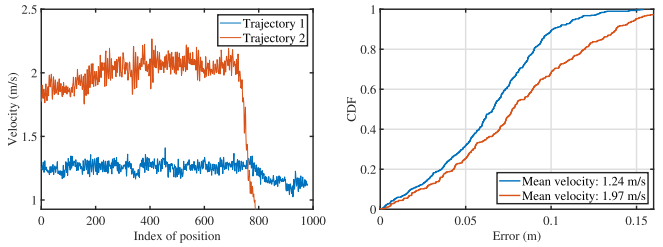


Fig. 25. (a) Velocity and (b) error distribution of two trajectories.

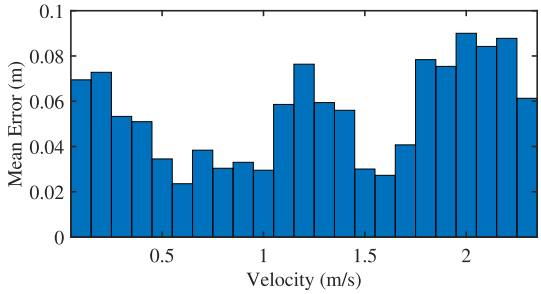


Fig. 26. Mean error in each velocity bin.

motion capture system where the target moved at different velocities. The average speed in the experiment shown in Fig. 15 was 0.44 m/s, while in the two experiments presented in Fig. 24, the average speeds were 1.24 and 1.97 m/s, respectively. Fig. 25(a) shows the recorded speed variations from the motion capture system, and Fig. 25(b) presents the cdf of localization errors from both trials, clearly indicating that lower speeds result in better accuracy. Combining data from these three experiments, Fig. 26 provides the average localization error for every 0.1-m/s velocity bin. Around 2 m/s, the average localization error approaches 0.08 m.

The breakdown of the localization accuracy at different average speeds is presented in Table VI. The initial trajectory had an average speed of 0.44 m/s. When the average speed increased by a factor of 2.82 to 1.24 m/s, the average error, mean error, and 90th percentile error increased by factors of 1.65, 2.10, and 1.23, respectively. At this point, the 90th percentile error reached 0.1009 m, meaning the system still maintained centimeter-level localization accuracy overall. When the average speed increased by a factor of 4.48 to 1.97 m/s, the average error, mean error, and 90th percentile error increased by factors of 2.08, 2.46, and 1.68, respectively. The average error is 0.0806 m, and the mean error is 0.0774 m, both still within the centimeter-level range. However, the 90th percentile error reaches 0.1379 m. As the speed continues to increase, the system's error is likely to grow further, potentially causing it to lose centimeter-level localization accuracy. This shows that the HARP system is primarily designed to achieve high update rates and does not incorporate Doppler compensation, which represents a direction for future improvement.

TABLE VI
STATISTICAL RESULTS OF DIFFERENT MEAN VELOCITIES

Error \ Mean Velocity	0.44 m/s	1.24 m/s	1.97 m/s
Mean (m)	0.0388	0.0640	0.0806
Median 50% (m)	0.0314	0.0660	0.0774
90% (m)	0.0819	0.1009	0.1379

In summary, challenging acoustic environments can degrade the system's localization performance. In future work, integrating corresponding algorithms could enhance the robustness of the system.

VII. CONCLUSION

This article thoroughly presents HARP, a 3-DoF acoustic localization system with a 50-Hz update rate. The system adopts a novel two-anchor layout, differing from traditional configurations, and incorporates an innovative signal scheme and detection method. Leveraging the combined localization strategy of TPC-RBL and CDI-RBL, high-precision RBL with only two anchors is achieved. Experimental results demonstrate that within a room-scale area, HARP achieves an average localization and angular error of just 0.0388 m and 6.4211°, respectively, meeting the demands of various mobile target localization and navigation applications.

HARP introduces a fundamentally different signal and localization scheme compared to previous approaches. Despite its significant breakthroughs, there remains room for improvement. First, the application of filtering methods, such as extended Kalman filtering, may further enhance accuracy. Second, with a high update rate, integration with other sensors like inertial measurement units (IMUs) could become more feasible and effective. Third, expanding the localization area by adding more anchors would differ from traditional system expansion methods, and further refinements in this aspect are needed. Lastly, to address challenges such as noise, multipath effects, and Doppler shifts in acoustic localization, integrating appropriate algorithms will help enhance the system's robustness.

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